REV.01

adaro
SERVICES
PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYLINDER HEAD

ASSEMBLY CHECKSHEET

KOMATSU ENGINE SAA6D170E-5

MACHINE MODEL

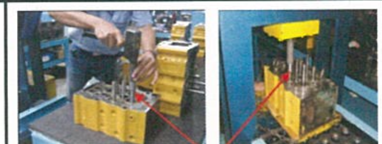
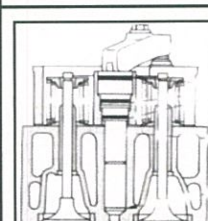
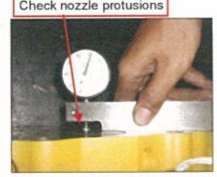
WO NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

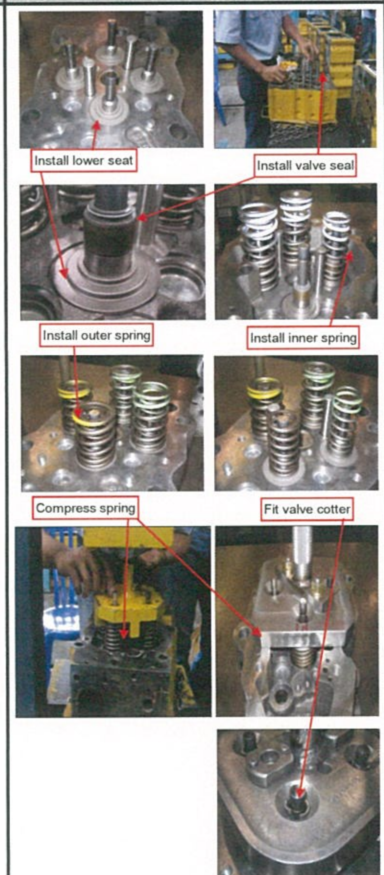
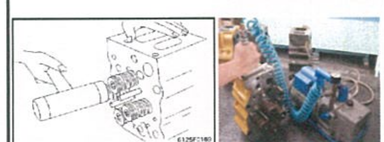
ASSEMBLY CHECK SHEET **CYLINDER HEAD** ENGINE ALL SERIES						
	WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	RC/QR04/EG/03/02
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	13/10/2009
	ENGINE S/N		START DATE /TIME		REVISION NO.	00
	MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

ASSEMBLY BY		All dimensions in millimetres All torques in Kilogram metres

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES & COMMENT)
1	CYL.HEAD ASSEMBLY	Head Assy Threads Cyl.Heads	- Head Assy must be free from crack, dents, pitting, rust and worn - Make sure all threads are not damage - Make sure top surface and bottom surface of the Cylinder heads are good conditions - Make sure the Cylinder Heads are clean					
2	VALVE GUIDES	Bores	- The Valve Guides be renew after measured. - Make sure the Valve Guide bores are clean when reinstall of the Valve Guides • Before inserting the valve guide, coat the outer dimension of the valve guide with small quantity of engine oil (SAE 15W-40)	GO and NO GO Hyd. Press				 Insert and press valve guides
3	INSERT VALVE	Insert Valve Bores Valve Sinking	- Make sure the dimension of the Insert Valve bores - Check that there is no dust, crack, corrosive and dirt - Install the Insert Valve be carefully • Diameter between Insert Valve bore and Insert Valve must interference see measurement check sheet • When install Insert Valves must straight to hole position - Check Valve sinking/Protrusion and refer to Standard (Shop Manual or measurement Check Sheet) • If sinking or Protrusion out of standard, cutting Insert or Valve and recheck sinking or Protrusion it must standard condition.	Special Tool Grinding Valve machine or Cutting Insert machine				 Install insert valve  Check valve sinking 
4	NOZZLE SLEEVE	Nozzle Sleeve Condition	- The Nozzle Sleeves must be renew - Check the O-ring grooves and the Nozzle Sleeve bores are clean when reinstall of the Nozzle Sleeves - Make sure the Nozzle Sleeves are in good position • When install and cutting the Nozzle Sleeves must use the Special Tool and refer to replacing Nozzle Holder Sleeve in Shop Manual (see replacing nozzle holder sleeve Job Instruction). • Check and measure protrusions of Nozzles after cutting Nozzle Sleeves (See and Record at the Measurement Check Sheet).					 Check nozzle protusions 
5	CRACK & LEAK TEST CYL.HEAD	Pressure test	- Air Pressure Test 1. Assemble plate and flange 2. Connect a hose to flange 3. Place the Cyl. Head in a water bath with hot water 4. Apply air pressure (3.0 ~ 3.5 kg/cm^2) for approx 5 ~ 10 Min. 5. Check for any air leakage in the water. 6. Replace Cyl. Head if there are crack.	3.0 ~ 3.5 kg/cm^2 for approx 5 ~ 10 Min. (82°C ~ 93°C)				 Check cylinder head with hot water & air pressure

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET **CYLINDER HEAD** ENGINE ALL SERIES	WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	RC/QR04/EG/03/02
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	13/10/2009
	ENGINE S/N		START DATE /TIME		REVISION NO.	00
	MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES & COMMENT)
6	VALVE & SPRING ASSEMBLY		- Make sure Cylinder Head free from Dirt and foreign material - Make sure all component that will assembly already clean 1. Align with casting mark on Cylinder Head and assembly Intake valve and Exhaust valve coat the valve steam and the face of the Valve thinly with engine oil (SAE 15W ~ 40) • Make sure of Valve (Intake and Exhaust) correct arrangement • Make sure the Valves are following to PSN Valve code					
		PSN	(see PSN table below)					
			2. Install lower seat to Cylinder Head, install valve seal (125 series + 140 series), then Assembly inner and outer spring 3. Install upper seat 4. Compress spring and fit Valve cotter in groove of valve steam • Make sure the Valve cotter good position in groove of Valve Steam • Recheck and make sure that Valve, Spring Valve and Seat Spring correct arrangement • Recheck pitch of Valve Spring (The Spring with the unequal pitch is assembled with the narrow pitch at the Head End) Tap Valve Steam end with plastic Hammer and check that Valve cotter is completely fitted. 5. Recheck Cylinder Head Assy completely. • Coat Cylinder Head (bottom surface, upper surface, Intake and Exhaust side and Spring) with engine oil SAE 15W - 40) • Plugging and masking the Cylinder Head. 6. Keep Cylinder Head Assy in keeping area.					 

PSN table:

ENGINE MODEL	VALVE INTAKE	VALVE EXHAUST
140 SERIES	I	CX / PX
170 SERIES	AN / BN	EX / X
125 SERIES	IN / BN	EX / BX
155 SERIES	I	E
170-3	IN	E
ACTUAL		

INSPECTION RESULT :

ASSEMBLY BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

PARTNER IN YOUR
MINING NEEDS

PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYLINDER HEAD

MEASUREMENT BEFORE MACHINING CHECK SHEET

ENGINE KOMATSU

6D170-5

MACHINE MODEL

WO NO:

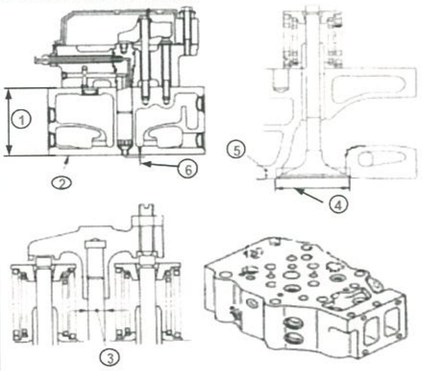
SUB ASSY SECTION

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

JOB NUMBER		**PART NAME**	CYLINDER HEAD	**DOCUMENT NO.**	RC/QA/EG/423
ENGINE MODEL	6D170E-5	**PART NUMBER**		**PUBL. DATE**	
ENGINE S/N		**START DATE / TIME**		**REVISION NO.**	4
MACHINE MODEL		**FINISH DATE / TIME**		**PAGE**	

MEASURED BY		**QUANTITY**	

STANDARD FIGURES



No	Measuring items			Std Tolerance (mm)	Repair Limit (mm)
1	Head Height			151.05 ~ 150.95	150,65
2	Lower Surface Distortion			Max. 0.05	0,10
3	O.D. of Cross Head Guide			13.039 ~ 13.028	12.999 and less
4	Valve Seat Insert Bore	In	Std	64.000 ~ 64.019	-
			0.25 OS	64.250 ~ 64.269	-
			0.50 OS	64.500 ~ 64.519	-
		Ex	Std	63.600 ~ 63.619	-
			0.25 OS	63.850 ~ 63.869	-
			0.50 OS	64.100 ~ 64.119	-
5	Valve Sink			0	0.80
6	Nozzle Protrusion			2.30 ~ 2.90	Replace Nozzle sleeve

A.Result of Measurement

No.	Measuring Item		1		2		3		4		5		6		
			MEC	QA	MEC	QA	MEC	QA	MEC	QA	MEC	QA	MEC	QA	
1	Head Height														
2	Distortion														
3	O.D Cross Head Guide	In													
		Ex													
4	Production Code														

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

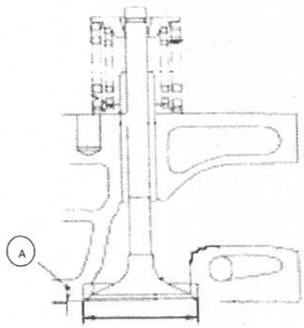
GL/SPV PRODUCTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER HEAD SALVAGING AND MEASUREMENT CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0023
ENGINE MODEL	ALL SERIES	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



Standard size

ENGINE MODEL	VALVE SINK(A) STANDARD(mm)	VALVE SINK(A) REPAIR LIMIT(mm)	VACUM TEST (BAR)
12V140	0 ± 0,10	0,70	- (6 ~ 8)
6D170-5	0± 0,10	0,80	
6D170E-3	0± 0,10	0,40	
6D140E-1	0 ± 0,10	0,70	
6D125-1	0 ± 0,10	0,70	

MAN HOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Salvaging			
Measuring			
TOTAL			

VISUAL CHECK	MECHANIC OK	MECHANIC NOT OK	QA OK	QA NOT OK
CRACK HAS BEEN CHECK				

No.	Measuring Item			R1 MEC	R1 QA/GL	R2 MEC	R2 QA/GL	R3 MEC	R3 QA/GL	R4 MEC	R4 QA/GL	R5 MEC	R5 QA/GL	R6 MEC	R6 QA/GL	RANDOM INSPECTION GL/SPV SIGN
1	Valve Sink	In	1													
			2													
		Ex	1													
			2													
2	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													

No.	Measuring Item			L1 MEC	L1 QA/GL	L2 MEC	L2 QA/GL	L3 MEC	L3 QA/GL	L4 MEC	L4 QA/GL	L5 MEC	L5 QA/GL	L6 MEC	L6 QA/GL	RANDOM INSPECTION GL/SPV SIGN
3	Valve Sink	In	1													
			2													
		Ex	1													
			2													
4	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													

INSPECTION RESULT :

DELIVERY BY MACHINING
DELIVERY DATE :
MACHINIST NAME :
SIGN

RECEIVING INSPECTION BY SUB ASSY
RECEIVED DATE :
MECHANIC NAME :
SIGN

Note : New Cyl Head & after finishing can be directly approval by QA (Quality Assurance)

PARTNER IN YOUR MINING NEEDS

PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYLINDER HEAD MEASUREMENT AFTER MACHINING CHECK SHEET

ENGINE KOMATSU

6D170-5

MACHINE MODEL

WO NO:

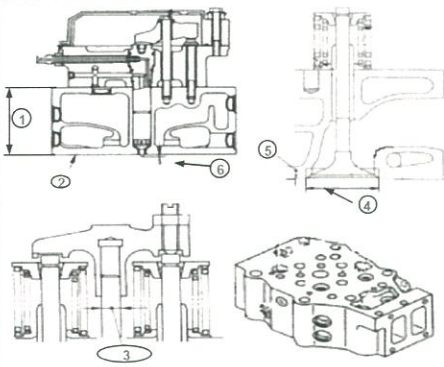
SUB ASSY SECTION

CYLINDER HEAD AFTER MACHINING AND MEASUREMENT CHECK SHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC/QA/EG/423
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	4
MACHINE MODEL		FINISH DATE / TIME		PAGE	

MEASURED BY		QUANTITY	

STANDARD FIGURES



No	Measuring items			Std Tolerance (mm)	Repair Limit (mm)
1	Head Height			151.05 ~ 150.95	150,65
2	Lower Surface Distortion			Max. 0.05	0,10
3	O.D. of Cross Head Guide			13.039 ~ 13.028	12.999 and less
4	Valve Seat Insert Bore	In	Std	64.000 ~ 64.019	-
			0.25 OS	64.250 ~ 64.269	-
			0.50 OS	64.500 ~ 64.519	-
		Ex	Std	63.600 ~ 63.619	-
			0.25 OS	63.850 ~ 63.869	-
			0.50 OS	64.100 ~ 64.119	-
5	Valve Sink			0	0.80
6	Nozzle Protrusion			2.30 ~ 2.90	Replace Nozzle sleeve

A.Result of Measurement

No.	Measuring Item			1		2		3		4		5		6		
				MEC	QA	MEC	QA	MEC	QA	MEC	QA	MEC	QA	MEC	QA	
1	Head Height															
2	Distortion															
3	O.D Cross Head Guide	In														
		Ex														
4	Valve Seat Insert Bore	In	1	MEASURED BY DIS-ASSEMBLY SECTION												
			2													
		Ex	1													
			2													
5	Valve Sink	In	1													
			2													
		Ex	1													
			2													
6	Nozzle protrution															
7	ID Valve Guide (Use Tool G/NG)	In	1	NEW PARTS												
			2													
		Ex	1													
			2													

No.	Inspection Item			R1		R2		R3		R4		R5		R6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	GL/SPV SIGN
8	Leaking Test Using Pressurized Air Method in Hot Water															
9	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													
10	Production Code															

Unit : mm

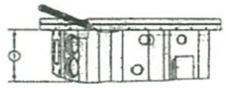
ENGINE MODEL	VALVE SPRING	STANDARD SIZE		REPAIR LIMIT
		FREE LENGTH	INSTAL LENGTH	LOAD (KG)
12V140 Series	INNER	72,00	46,00	19,70
	OUTER	81,70	51,00	38,70
6D140 Series	INNER	72,00	46,00	19,60
	OUTER	81,70	51,00	38,50
6D170-3	INNER	-	-	-
	OUTER	99,60	87,40	62,00
6D170 Series	INNER	59,30	47,00	17,40
	OUTER	64,80	37,50	84,80
6D170 HD465-5	INNER	73,80	48,00	23,30
	OUTER	77,00	52,00	48,70
6D125-2	INNER	-	-	-
	OUTER	75,20	56,00	47,70
6D155-4	INNER	-	-	-
	OUTER	88,00	51,90	65,50

VALVE SPRING			NUMBER OF CYL HEAD											
			1		2		3		4		5		6	
			FREE LENGTH	LOAD	FREE LENGTH	LOAD	FREE LENGTH	LOAD	FREE LENGTH	LOAD	FREE LENGTH	LOAD	FREE LENGTH	LOAD
IN	1	INNER												
		OUTER												
	2	INNER												
		OUTER												
EX	1	INNER												
		OUTER												
	2	INNER												
		OUTER												
IN	1	INNER												
		OUTER												
	2	INNER												
		OUTER												
EX	1	INNER												
		OUTER												
	2	INNER												
		OUTER												

Measurement procedure of flatness Cylinder head surface

Method of measurement of depth, and measurement tool :
Visual inspection or use Dial gauge.

Judgement of Distortion(Flatness)
Measure clearance between Head and straight edge
Measuring point : Refer the right picture



Upper surface	Flatness	Remedy
	within 0.08mm	Use as it is
	More than 0.08mm	Reuse after surface grinding
lower surface	within 0.08mm	Reuse after surface grinding

Measurement of Distortion , and measurement tool
aight edge, Clearance gauge

Measure flatness of the bottom surface of head.

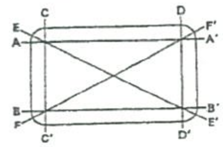


Fig. A

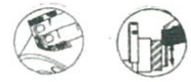


Fig. B

INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION

REV.01

adaro SERVICES
PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CYLINDER HEAD

SALVAGING AND MEASUREMENT CHECKSHEET

KOMATSU ENGINE SAA6D170E-5

MACHINE MODEL

WO NO:

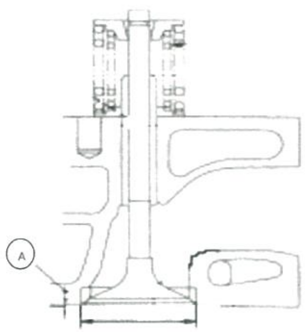
SUB ASSY SECTION

adaro SERVICES
PT SAPTAINDRA SEJATI

CYLINDER HEAD SALVAGING AND MEASUREMENT CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0023
ENGINE MODEL	ALL SERIES	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



Standard size

ENGINE	VALVE SINK(A)		VACUM TEST
MODEL	STANDARD(mm)	REPAIR LIMIT(mm)	(BAR)
12V140	0 ± 0,10	0,70	- (6 ~ 8)
6D170-5	0± 0,10	0,80	
6D170E-3	0± 0,10	0,40	
6D140E-1	0 ± 0,10	0,70	
6D125-1	0 ± 0,10	0,70	

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Salvaging			
Measuring			
TOTAL			

VISUAL CHECK	MECHANIC		QA	
	OK	NOT OK	OK	NOT OK
CRACK HAS BEEN CHECK				

No.	Measuring Item			R1		R2		R3		R4		R5		R6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	GL/SPV SIGN
1	Valve Sink	In	1													
			2													
		Ex	1													
			2													
2	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													
	Measuring Item			L1		L2		L3		L4		L5		L6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	GL/SPV SIGN
3	Valve Sink	In	1													
			2													
		Ex	1													
			2													
4	Vacuum Test	STD	IN													
		(6~8 Bar)	EX													

INSPECTION RESULT :

DELIVERY BY MACHINING
DELIVERY DATE :
MACHINIST NAME :
SIGN

RECEIVING INSPECTION BY SUB ASSY
RECEIVED DATE :
MECHANIC NAME :
SIGN

Note : New Cyl Head & after finishing can be directly approval by QA (Quality Assurance)

RC/QA/EG/436	REV. 04

PARTNER IN YOUR MINING NEEDS

PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

IDLER GEAR AND SHAFT MEASURING AND INSPECTION CHECK SHEET

KOMATSU ENGINE 6D170-5

MACHINE MODEL

WO :

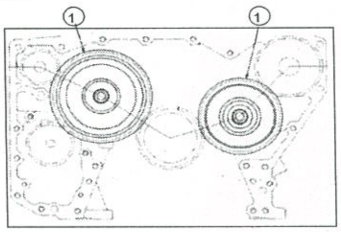
SUB ASSY SECTION

MEASURING IDLER GEAR AND SHAFT
INSPECTION CHECKSHEET

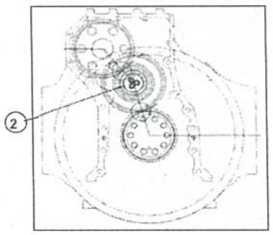
JOB NUMBER		PART NAME	IDLER GEAR AND SHAFT	DOCUMENT NO.	RC/QA/EG/436
ENGINE MODEL	6D170E-5 Series	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	4
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING

FRONT SIDE



REAR SIDE



Physical Data

No.	Part Number	Part Name	Qty	New	Reuse
1					
2					
3					
4					

MECHANIC NAME	

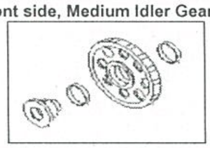
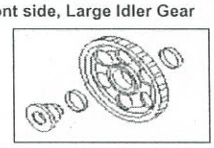
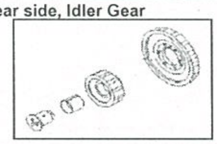
Standard size (unit mm)

DESCRIPTION	STANDARD DIMENSION		CLEARANCE	
	SHAFT	BUSHING	STANDARD	REPAIR LIMIT
1. Main Idler Gear (R and L Bank)	47.614 ~ 47.601	47.639 ~ 47.663	0.025 ~ 0.062	0.25
2. Sub Idler Gear	51.973 ~ 51.960	52.000 ~ 52.055	0.025 ~ 0.062	0.25
3.Distance betwen bushing to bushing on sub idle gear.	STANDART DISTANCE			
	3.8			

VISUAL CHECK	MECHANIC		QA	
	OK	NOT	OK	NOT
CRACK HAS BEEN CHECKED				

MANHOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Machining			
Measuring			
TOTAL			

INSPECTION SALVAGE PROCESS

SKETCH DRAWING	DIMENSION		ACTUAL DIAMETER						ACTUAL CLEARANCE			ROTATING CHECK (OK or NOT)		
			SHAFT			BUSHING								
			MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV
Front side, Medium Idler Gear 	A	X							MIN					
		Y												
	B	X												
		Y							MAX					
	C	X												
		Y												
Front side, Large Idler Gear 	A	X							MIN					
		Y												
	B	X												
		Y							MAX					
	C	X												
		Y												
Rear side, Idler Gear 	A	X							MIN					
		Y												
	B	X												
		Y							MAX					
	C	X												
		Y												

MECHANIC INSPECTION COMMENT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTON

REV.01

adaro
SERVICES
PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ROCKER ARM SHAFT

SALVAGING AND MEASUREMENT & POLISHING CHECKSHEET

KOMATSU ENGINE SAA6D170E-5

MACHINE MODEL

WO NO:

SUB ASSY SECTION

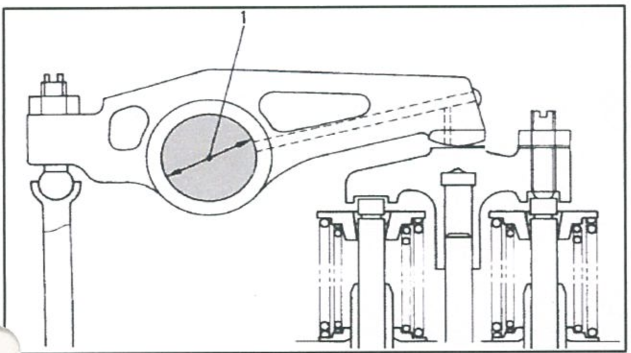
PARTNER IN YOUR MINING NEEDS

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ROCKER ARM SHAFT
MEASURING AND POLISHING CHECK SHEET

WO NUMBER		PART NAME	ROCKER ARM SHAFT	DOCUMENT NO.	RC/QR03/EG/09/03
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	05/02/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



MACHINIST NAME	

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Machining			
Measuring			
TOTAL			

(Unit : mm)

ENGINE MODEL	STANDARD	STANDARD TOLERANCE	CLEARANCE LIMIT
6D170E-5	ø 45.0000	ø 44,985~44,972	0,16

INSPECTION MEASURING PROCESS

CYL.NO.	CHECK POSITION	1			2			3			Actual Clearance
		MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	
1	X – X'										
	Y – Y'										
2	X – X'										
	Y – Y'										
3	X – X'										
	Y – Y'										
4	X – X'										
	Y – Y'										
5	X – X'										
	Y – Y'										
6	X – X'										
	Y – Y'										

MACHINIST INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV.01

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ROCKER ARM

POLISHING & INSPECTION CHECKSHEET

KOMATSU ENGINE SAA6D170E-5

MACHINE MODEL

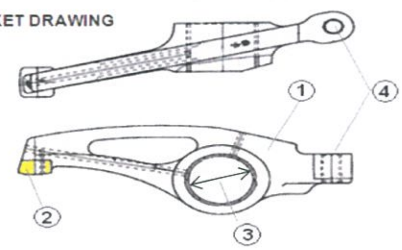
WO NO:

SUB ASSY SECTION

ROCKER ARM POLISHING
INSPECTION CHECK SHEET

JOB NUMBER		PART NAME	ROCKER ARM	DOCUMENT NO.	RC/QR03/EG/09/03
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	16/03/2009
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKET DRAWING



MACHINIST NAME	

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Receiving Inspection			
TOTAL			

Physical Data

No	Part Name	Part Number	Qty	New	Reuse
1	Cam follower				

INSPECTION OF ROCKER ARM √= Good X = No Good

No	Check Point	Check Content	Standard Reuseble Parts	Methode	1		2		3		4		5		6		Remarks
					IN	EX	IN	EX	IN	EX	IN	EX	IN	EX	IN	EX	
1	Arm assy	Scracthes	No scratch	Visual check													
		Appearance	Rivet no come out	Visual check													
		Oil hole	No clog	Visual check													
2	Contact surface	Scracthes	No scratch	Visual check													
		Crack	No crack	Visual check													
		Wear	No stepped type wear	Visual check													
3	Bush inside	Scrateches	No scratch	Visual check													
		Wear	Max. mm	Visual check													
4	Screw Portion	Wear	No wear	Visual check													

RECEIVING INSPECTION COMMENT :

DELIVERY BY MACHINING
DELIVERED DATE :
MACHINIST NAME :
SIGN

INSPECTION BY QUALITY ASSURANCE
DATE :
MECHANIC NAME :
SIGN

Note : New Rocker Arm & after finishing can be directly measured by QA (Quality Assurance)

QR03-SA-008-03	REV.01

PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

TENSION PULLEY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5 SERIES

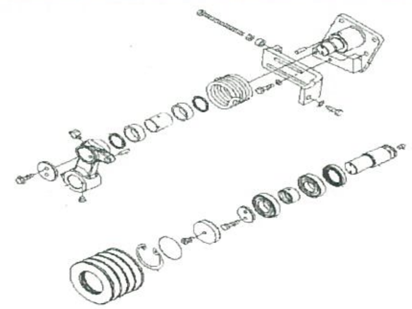
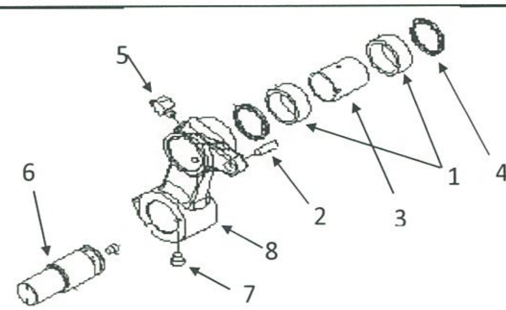
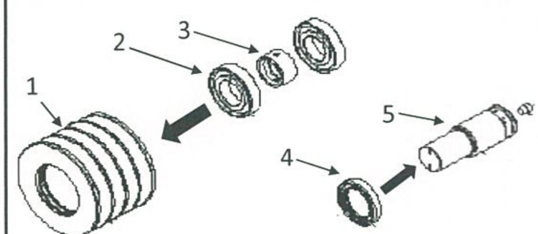
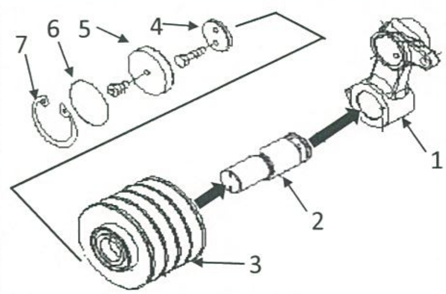
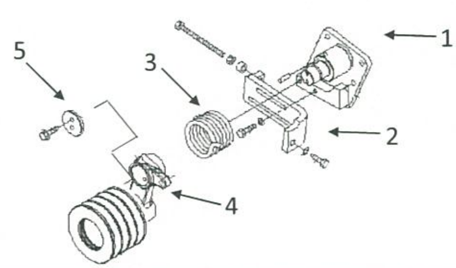
MACHINE MODEL

WO NO:

SUB ASSY SECTION

ASSEMBLY CHECK SHEET **TENSION PULLEY** 6D170-5 SERIES			DOCUMENT NO	RC/QR04/EG/12/03
	WO NUMBER		DOCUMENT NO	RC/QR04/EG/12/03
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START		ASSEMBLY BY		
	FINISH				

NO	PART TO INSTALL OR FIT	CONDITION TOBE ASSAMBLY	SPECIFICATION	SIGNATURE MECH	SIGNATURE GL/QA	SKETCH DRAWING OR PHOTHOES
1	ALL PART	**⚠ IMPORTANT ⚠** Confirm that Parts which have order do not DAMAGE,SCRATCH FREE FROM DIRT AND COROTION.				
2	TENSION LEVER	• Make sure tension lever(8) is clean and free from damage. • Reamer bearing seat hole then install inner race(3) and neddle bearing(1). • Install dust seal (4). • Install shaft(6) to lever tension pulley(8). • Insert plug(7),dowel(2) and elbow(5)	Using special tool for press (F2) Using special tool Using special tool for press (F2)			
3	PULLEY ASSEMBLY	• Make sure pulley is clean and free from damage. • Install oil seal (4) to the shaft. • Install bearing(2) and the spacher(3) to the pulley(1).	Using special tool. Using special tool for press (F2)			
4	PULLEY,SHAFT AND LEVER ASSY	• Install shaft assy(2) to tension lever assy(1). • Install Pulley assy(3) to shaft assy(2). • Install Cover lock(4) ,then tighten mounting bolt. Actual= Kgm • Install cover(5),o ring(6) , snapring(7) and plug.	Using press tool 3,2 ± 0,3 Kgm			
5	ASSEMBLY LEVER ASSY AND TENSION SHAFT	• Install bracket(2),spring(3) and shaft assy(1). • Mounting barcket(2) to shaft(1) and tighten. • Install lever assy(4) to shaft(1). • Install cover (5),mounting bolt and tighten. Actual= Kgm	6,7 ± 0,7 Kgm			

ASSEMBLY CHECK SHEET **TENSION PULLEY** 6D170-5 SERIES	WO NUMBER		DOCUMENT NO	RC/QR04/EG/12/03
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

6	FITTING GREASE AND PLUG	• Install fitting grease and plug ,then tighten. • Give all surface pulley with grease to protect from rust.				

NOTE :

3

ASSEMBLY BY	INSPECTED BY	APPROVED BY
MECHANIC	Quallity Assurance(QA)	GL/SPV PRODUCTION

QR03-SA-008-03	REV.01

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PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET KOMATSU ENGINE SAA6D170E-5

MACHINE MODEL

WO NO:

SUB ASSY SECTION

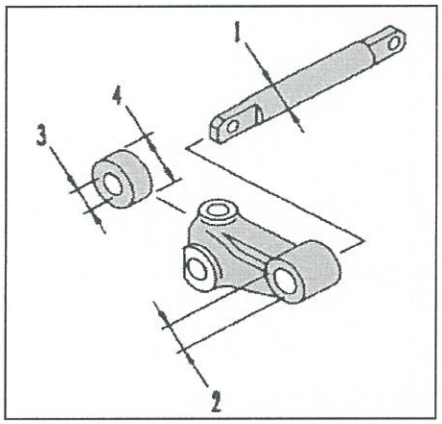
PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

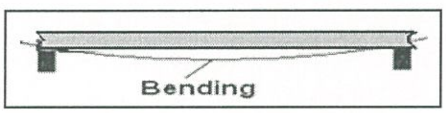
CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	Eks UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	28/05/2010
ENGINE S/N		START DATE/TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE/TIME		DOCUMENT NO.	1 of 4

MEASURED BY	

STANDARD FIGURE





Unit : mm

ENGINE MODEL	CAM FOLLOWER RADIAL PLAY	
	STANDARD	Repair Limit
6D170E-5	0.082 ~ 0.150	0.151

Unit : mm

PUSH ROD BENDING STANDARD	
TIR max.	0.30

6D170E-3

Unit : mm

NO	CHECK ITEM	DIMENSION		
		SIZE	STD TOLERANCE	Repair Limit
1	OD of C.F. Shaft	25	24.947 ~ 24.962	25
	ID of C.F. Shaft Hole	25	25.000 ~ 25.021	-
2	OD Cam Roller	50	50.025 ~ 50.000	49.75

RADIAL PLAY CAM FOLLOWER						
CAM FOLLOWER	IN			EX		
	MECHANIC	QA/GL	SPV	MECHANIC	QA/GL	SPV
1						
2						
3						
4						
5						
6						

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	Eks UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	28/05/2010
ENGINE S/N		START DATE/TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE/TIME		DOCUMENT NO.	2 of 4

BENDING OF PUSH ROD						
PUSH ROD	IN			EX		
	MECHANIC	QA/GL	SPV	MECHANIC	QA/GL	SPV
1						
2						
3						
4						
5						
6						

OUTSIDE DIAMETER OF CAM FOLLOWER SHAFT								
Out Side Diameter of Cam Follower Shaft & Hole	MECHANIC		QA/GL	SPV	MECHANIC		QA/GL	SPV
	X	X'			Y	Y'		
1								
2								
3								
4								
5								
6								

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	Eks UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	28/05/2010
ENGINE S/N		START DATE/TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE/TIME		DOCUMENT NO.	3 of 4

INSIDE DIAMETER OF CAM FOLLOWER HOLE						
Inside Diameter of Cam Follower Shaft & hole	X			Y		
	MECHANIC	QA/GL	SPV	MECHANIC	QA/GL	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

OUSIDE DIAMETER OF CAM ROLLER						
Outside Diameter Of Cam Roller	X			Y		
	MECHANIC	QA/GL	SPV	MECHANIC	QA/GL	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	Eks UNIT CODE	
ENGINE MODEL	6D170E-5	PART NUMBER		PUBL. DATE	28/05/2010
ENGINE S/N		START DATE/TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE/TIME		DOCUMENT NO.	4 of 4

MECHANIC INSPECTION COMMENT :

INSPECTED BY :

INSPECTED & APPROVED BY :

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

REV.01

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SERVICES
PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

WATER PUMP ASSY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

JOB NO:

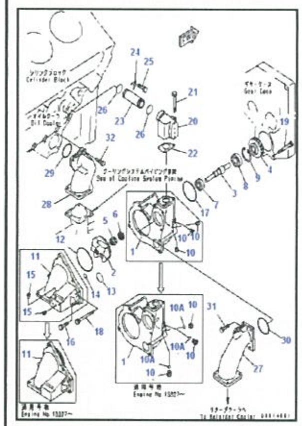
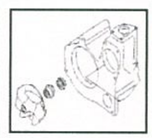
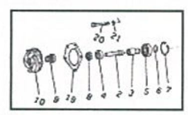
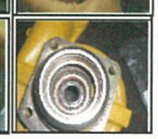
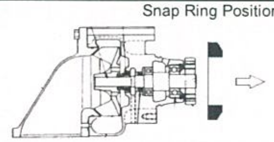
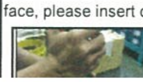
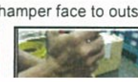
SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
WATER PUMP
6D170&12V140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QR04/EG/04/02
ENGINE MODEL		PART NUMBER		PUBLIC.DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	2
MACHINE MODEL		FINISH DATE / TIME		PAGE	

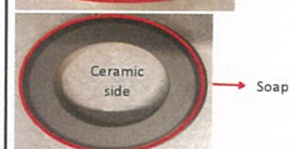
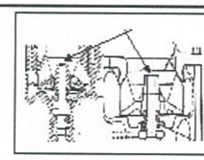
ASSEMBLY BY		All dimension in millimetres All torques in Kilogram metres

NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA/GL	REMARKS (SKETCH, PHOTOES, COMMENTS)
1	Pump Body		Pump body must be free from crack, dent and rush Clean All Parts throughly and make sure that all burns have been remove. Always check the contact of the following parts before assembly. See checking contact between parts. Marking with paint maker, for all of bolts and nuts after tightening by torque wrench	See maintenance standard for standard dimension			
2	Housing		Housing must be free from crack, dents and rust All threads are in good condition and clean Make sure that face (clearance area) is in good and clean				 
3	Other Parts		Recheck condition of bearings, oil seal, shaft, impeller, water seal, gear, snap ring, bolts and washers should be free from crack, rust, dents, scratch and pitting	Refer and follow PSN			There are many PSN for the Water pump 
4	Insertion of Oil Seal		Oil seal seat must be clean and dry (free from oil, fuel or water) Coat the out side of oil seal with Red Liquid gasket. Insert oil seal using special tools Coat the seal lip with "grease" sufficiently	Use oil seal installer **LG 1207** **G2-LI**			   
5	Insertion Bearing to Shaft		Insert small bearing and big bearing to shaft using special tools Make sure that bearings center to shaft when assembly **For 12V140 Series** Before inserting the small bearing and the big bearing, make sure that the shaft is new shaft with model P/n.6219-61-1230 (shaft end stamping reading "S5")	Use special Tools and Press by Hyd.Jack			   Make sure that bearings center to the shaft when assembly
6	Shaft Assembly		Coat the shaft (seal contact) by sufficient grease Bearings and bearing seat must be clean Insert shaft ass'y to the body and make sure that shaft assy is centering to body CAUTION Recheck and make sure that oil seal isn'n hit or damage when inserting the shaft Check shaft from smooth rotation	Use special Tools (Shaft Installer)			Snap Ring Position  Big Snap ring (at water pump) has different face, please insert champer face to outside.  

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
WATER PUMP
6D170&12V140 SERIES

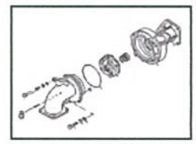
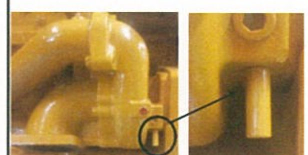
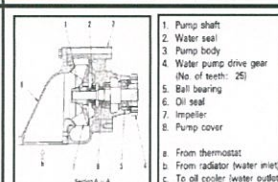
WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QR04/EG/04/02
ENGINE MODEL		PART NUMBER		PUBLIC.DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	2
MACHINE MODEL		FINISH DATE / TIME		PAGE	

NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA/GL	REMARKS (SKETCH, PHOTOES, COMMENTS)
7	Insertion of Drive Gear		Shaft and gear contact must be clean Coat the drive gear hole with engine oil Install Spacer. Insert drive gear to the shaft until contact to inner race bearing or stopper Standard of interference between Gear and Shaft **Act. Interference : mm**	Drive Gear Installer **0.02 ~ 0.06 mm**			When inserting the drive gear Please arrest the shaft at the End of the shaft  
8	Insertion of Water Seal		Make sure that the condition of water seal is free from scratch, crack, pitting, dents, corrosive, worn, and etc. **Here is critical point, be careful** Coat the outer side of water seal with adhesive Insert water seal to housing with special tool. Coat Ceramic surface of water seal with **"toilet soap"**	Follow and refer to the PSN **LOCTITE 609 (green loctite)** **TOILET SOAP**			 
9	Insertion of Impeller		Insert water seal using special tools Insert the seal ring to the groove at impeller and make sure that it is fit in the groove Coat the hole of the impeller with small quantity of **Engine Oil** be carefull when insert the Impeller	**SAE 15W-40**			 
			For 6D170 Series When inserting the impeller , adjust impeller from end of shaft using special tool (spacer) to assure Standard of Spacer Thickness **Act. Spacer thickness = mm** Clearence between impeller and connector / housing Standard Clearance **Act. Clearance = mm** **For 12V140 Series** Before inserting the impeller, make sure that impeller is new impeller model with P/n.6219-61-1470 (change the hole for the shaft) **Act. Clearance = mm** Insert the impeller to the shaft until contact to the end of the shaft.	**1.2 ~ 1.7 mm** **0.23 ~ 0.66 mm** Standard Clearence : **0.25 ~1.02 mm**			  

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
WATER PUMP
6D170&12V140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QR04/EG/04/02
ENGINE MODEL		PART NUMBER		PUBLIC.DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	2
MACHINE MODEL		FINISH DATE / TIME		PAGE	

NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA/GL	REMARKS (SKETCH, PHOTOES, COMMENTS)
10	Insertion of Housing Hole Indicator		O-ring groove must be free from crack, scratch, pitting, dents and rust Check O-ring from scratch Insert O-ring and check from twist and out of it's groove Insert Housing/connection and tighten mounting bolts carefully and properly Act. Torque M8 = Kgm Act. Torque M10 = Kgm Act. Torque M12 = Kgm Install extension tube to hole indicator	Tightening torque : M8 = 3.2 + 0.3 kgm M10= 6.7 + 0.7 kgm M12= 11.5 + 1.0 kgm			    
11	General Inspection		√ : Check - Check Shaft for smooth rotation OK - Recheck position of snap ring OK - Check loose bolts from mistake OK - Check all parts from improper installation OK - Check and keep completion parts OK				 
12	End play water pump		Measure End play of water pump shaft before delivery to Main line	Standard End play 0			

MECHANIC ASSEMBLY COMMENT

ASSEMBLY BY		INSPECTED & APPROVED BY
MECHANIC (MECH)	QUALITY ASSURANCE (QA)	PROD. SPV/GL

REV.01

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

COMMON RAIL ASSY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

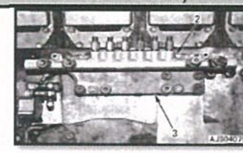
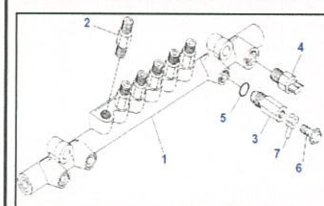
JOB NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CHECK SHEET COMMON RAIL ASSEMBLY KOMATSU ENGINE 6D170E-5				
	WO NUMBER		DOCUMENT NO	RC/QR04/EG/05/01
	ENGINE MODEL	SAA6D170E-5	PUBLICATION DATE	
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS			ASSEMBLY BY	
	START DATE / TIME			
	FINISH DATE / TIME			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
1	Common Rail Element	Housing	- Inspect Housing for cracks by visual check. No Crack [ ] - Before assembling ,wash & clean the Housing.	√= CHECK				
2	ALL PARTS	Flow Damper Plate,Plunger,Spring and Ball	- Make sure all Flow Damper are ready to used (free Crack ; worn; corrosion and dirt) -Make sure All inner Parts are ready to used (free Crack ; worn; corrosion and dirt) - Before assembling , wash and clean the parts. (Allow them to dry by air blow) - Specification of spare parts are correct	√= CHECK √= CHECK				     
3	PRESSURE LIMITER	Fuel Pressure Limiter and Plug	- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) Make sure Thread of Hexagon Plug front & rear and Thread of Pressure Limiter good condition(not broken)	√= CHECK				
4	Assembly Common Rail		- Coat Inside surface of Housing and Inner parts with engine oil (SAE # 30) -Install Plate to Main Housing -Install Plunger to spring then Install to Flow Damper. and Tighten Flow damper (**2**): Actual : kgm -Install Ball & Hexagon Plug on front & rear of Housing and tighten Plug use L key socket: *- Before tighten Plug to the Housing coat a thin of adhesive LT-2 to the thread of Plug(High streng type).* Actual : kgm -Install Fuel pressure Limiter (**3**) with Picco Ring (**5**) to Main housing and Tighten. Actual : kgm -Install Fuel pressure sensor (4). to Main housing and Tighten. Actual : kgm	13 ± 1.0 kgm (Reman Manual) 5 ± 0.5 kgm (Komatsu KES Plug Standard) 9 ± 0.5 kgm (Reman Manual) 5 Kgm (Reman Manual)				 

MECHANIC ASSEMBLY COMMENT :

ASSEMBLY BY

INSPECTION & APPROVED BY

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

REV.00

adaro SERVICES
PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

OIL COOLER FRONT ASSY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

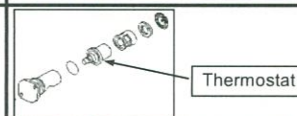
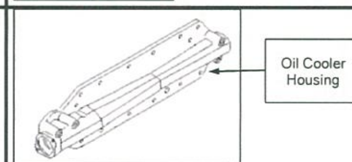
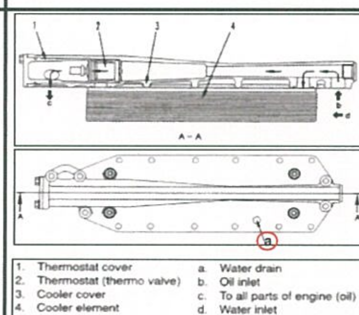
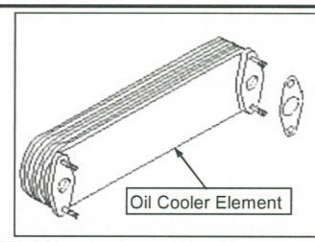
JOB NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET **OIL COOLER FRONT** KOMATSU ENGINE 6D170-5	WO NUMBER		DOCUMENT NO	
	ENGINE MODEL		PUBLISHED DATE	07/08/2009
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
					MEC	SPV	QA	
1	Core Element		• Test and check flow on core element. • Test and check leakage on core element	Oil cooler flusher				 
2	Thermostat		• Check cleanliness and damage crack on thermostat. • Test and check function of thermostat.					
3	Housing		• Check cleanliness and damage crack on element housing. • Check all threads on element housing • Check paper gasket threebond spec for sustainable genuine part. • Prepare liquid gasket threebond 1207C • Install o ring to plug and install to housing then tighten. Actual : Kgm	Red liquid gasket 5 Kgm				
4	Assembly of Thermostat		• Install seal on housing thermostat. • Install thermostat on housing oil cooler • Install o ring on cover thermostat • Install cover thermostat on oil cooler housing . • Install bolt then tighten. Actual : Kgm	3.2 ±0.3Kgm				
5	Assembly of Oil Cooler Element		• Fit O ring or gasket on housing • Install washer and nut then tighten. • Tighten Torque : M10 = Actual : Kgm M12 =	Apply LG-6 2.8~3.8Kgm 5.6~6.6Kgm (Komatsu assembly manual)				

Comment :

ASSEMBLY BY

INSPECTION & APPROVED BY

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

OIL COOLER REAR ASSY

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

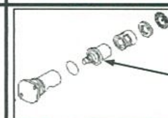
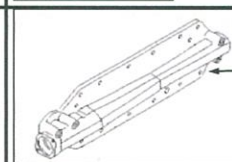
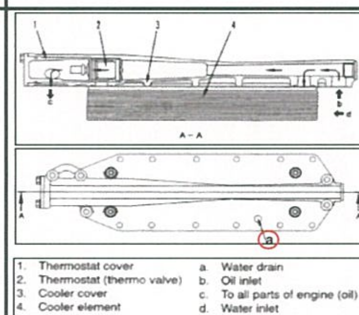
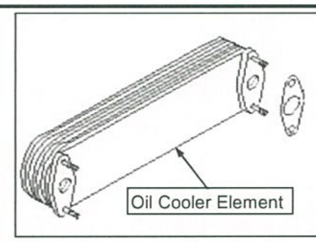
JOB NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET **OIL COOLER REAR** KOMATSU ENGINE 6D170-5				
	WO NUMBER		DOCUMENT NO	
	ENGINE MODEL		PUBLISHED DATE	07/08/2009
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
					MEC	SPV	QA	
1	Core Element		• Test and check flow on core element. • Test and check leakage on core element	Oil cooler flusher				 
2	Thermostat		• Check cleanliness and damage crack on thermostat. • Test and check function of thermostat.					 Thermostat
3	Housing		• Check cleanliness and damage crack on element housing. • Check all threads on element housing • Check paper gasket threebond spec for sustainable genuine part. • Prepare liquid gasket threebond 1207C • Install o ring to plug and install to housing then tighten. Actual : Kgm	Red liquid gasket **5 Kgm**				 Oil Cooler Housing
4	Assembly of Thermostat		• Install seal on housing thermostat. • Install thermostat on housing oil cooler • Install o ring on cover thermostat • Install cover thermostat on oil cooler housing . • Install bolt then tighten. Actual : Kgm	**3.2 ±0.3Kgm**				
5	Assembly of Oil Cooler Element		• Fit O ring or gasket on housing • Install washer and nut then tighten. • Tighten Torque : **M10 =** **M12 =** **Actual :** **Kgm**	**Apply LG-6** **2.8~3.8Kgm** **5.6~6.6Kgm** (Komatsu assembly manual)				

Comment :

ASSEMBLY BY

INSPECTION & APPROVED BY

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

OIL COOLER ASSY

AIR LEAK TEST CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

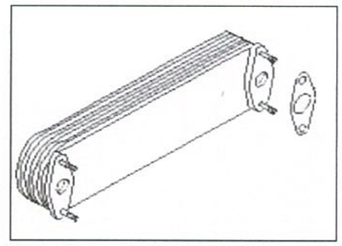
JOB NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

AIR LEAK TEST CHECK SHEET **OIL COOLER ASSY** **KOMATSU ENGINE 6D170-5**	JOB NUMBER		DOCUMENT NO	RC/QR05/EG/04/02
	ENGINE MODEL		PUBLISHED DATE	28/04/2009
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
					MEC	SPV	QA	
1	Oil Coler Element	Air Leak Tester	Inspect Element By Visual Check No Crack					 
2	Test Pressure		Check for Crack By Air Pressure Test For Element a).Air Pressure Actual: kgm/cm2 b).Water temperatur Actual: °C c).Time for Test Actual: minutes No Leakage					

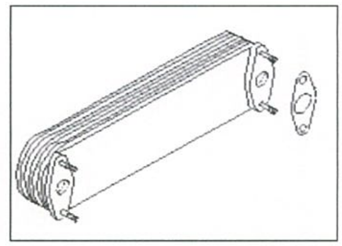
Comment :

ASSEMBLY BY	INSPECTION & APPROVED BY	
MECHANIC (MEC)	QUALITY ASSURANCE (QA)	PROD. SPV/GL

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

AIR LEAK TEST CHECK SHEET OIL COOLER ASSY KOMATSU ENGINE 6D170-5	JOB NUMBER		DOCUMENT NO	RC/QR05/EG/04/02
	ENGINE MODEL		PUBLISHED DATE	28/04/2009
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
					MEC	SPV	QA	
1	Oil Coler Element	Air Leak Tester	Inspect Element By Visual Check No Crack					 
2	Test Pressure		Check for Crack By Air Pressure Test For Element a).Air Pressure Actual: kgm/cm2 b).Water temperatur Actual: °C c).Time for Test Actual: minutes No Leakage					

Comment :

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

OIL COOLER ASSY

WATER LEAK TEST CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

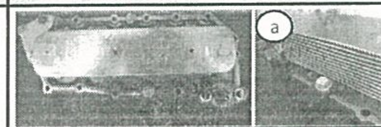
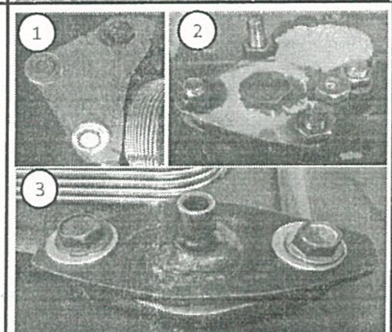
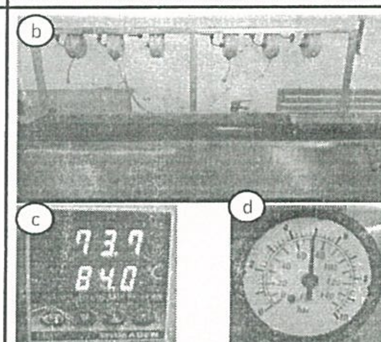
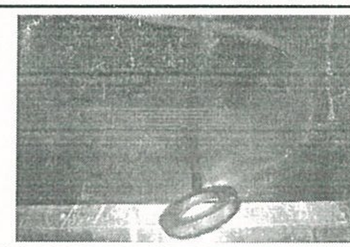
JOB NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

WATER LEAK TEST CHECK SHEET **OIL COOLER ASSY** KOMATSU ENGINE 6D170 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	6D170 SERIES	PUBLISHED DATE	22-Jan-15
	ENGINE S/N		REVISION NO	01
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START		TEST BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
1	Core Element		• Check Element normal condition. (broken from bend,rust etc) • Install hook/hanger (a).	Visual check				
2	Plug,adapter and Air inlet connector		• Install plug on housing (1). • Install adapter and use O ring (2) • Install inlet air connector and use sealant or rubber for this connector (3)					
3	Water leak tester and indicator		• Prepare water leak tester tool (b) • Setting water temperature at standar range '(c). Actual : °C • Setting air pressure on Regulator valve (d) Actual : Kg/cm	Water leak tester tool STD range tmpratur 75° ~ 90°C (Shop manual) 3 ~ 4 kg/cm (Reman manual)				
4	Water leak test oil cooler		• Install coupler and test oil cooler assy Time for test Actual : Minutes	5 Minutes (Reman manual)				

Comment :

ASSEMBLY BY

INSPECTION & APPROVED BY

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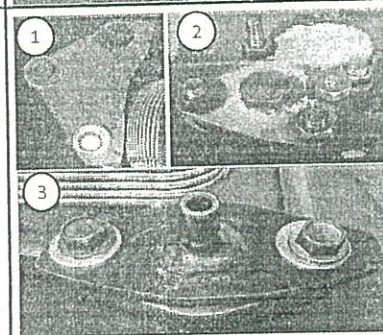
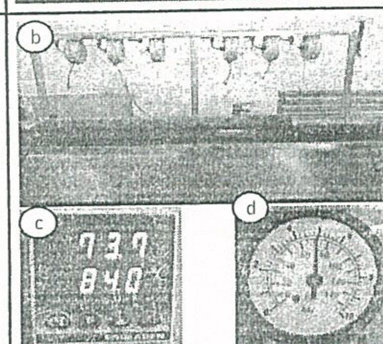
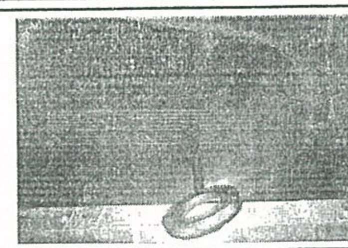
PROD. SUPERVISOR (SPV)

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

WATER LEAK TEST CHECK SHEET **OIL COOLER ASSY** KOMATSU ENGINE 6D170 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	6D170 SERIES	PUBLISHED DATE	22-Jan-15
	ENGINE S/N		REVISION NO	01
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START		TEST BY	
	FINISH			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
1	Core Element		• Check Element normal condition. (broken from bend,rust etc) • Install hook/hanger (a).	Visual check				
2	Plug,adapter and Air inlet connector		• Install plug on housing (1). • Install adapter and use O ring (2) • Install inlet air connector and use sealant or rubber for this connector (3)					
3	Water leak tester and indicator		• Prepare water leak tester tool (b) • Setting water temperature at standar range '(c). Actual : °C • Setting air pressure on Regulator valve (d) Actual : Kg/cm	Water leak tester tool STD range tmpratur 75° ~ 90°C (Shop manual) 3 ~ 4 kg/cm (Reman manual)				
4	Water leak test oil cooler		• Install coupler and test oil cooler assy Time for test Actual : Minutes	5 Minutes (Reman manual)				

Comment :

ASSEMBLY BY		INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA)	PROD. SUPERVISOR (SPV)

REV.02

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PT. SAPTAINDRA SEJATI

PLANT REBUILD CENTER

TURBO CHARGER

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

JOB NO:

SUB ASSY SECTION

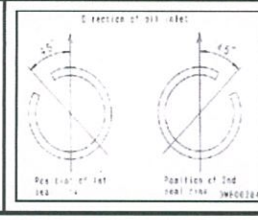
PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D170 SERIES						
	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

ASSEMBLY BY		All dimensions in millimetres
		All torques in Kilogram metres

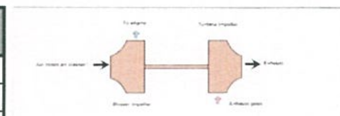
Notes :

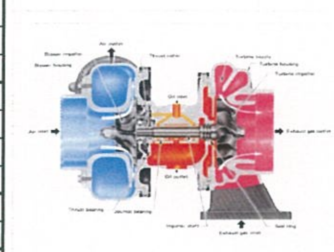
- Before assembly parts,wash and clean the parts,allow them to dry by air blow,and apply engine oil to the sliding parts.
- Remove burrs completely from the parts (Pay attention to seal ring grooves,in particular).
- When reassembling the rotor assembly after the dynamic balance check has been made, be sure to align all the match marks stamped on the component parts (Turbine Shaft ; Nut ; Blower Impeller ; Flinger and Collar).

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	NOTES, DIAGRAMS COMMENTS
1	Turbine Rotor	Seal ring Groove	Fit seal ring into Turbine rotor. Apply a coat of engine oil (SAE #30) to the side of seal ring and install the seal ring in such a way that the gap of each ring is positioned at 90° apart from that of the neighboring ring or 45° apart from oil filler port on the center housing.					

Thickness of Seal Ring

	Standard size		Tolerance Width	Repair Limit Width	Actual
KTR 100	Turbine Side	2,2	-0,08	1,85	
			-0,1		
	Blower Side	1,95	-0,08	1,85	
			-0,1		
		4	-	-	
KTR 110	Turbine Side	2,2	-0,08	1,85	
			-0,1		
	Blower Side	1,95	-0,08	1,45	
			-0,1		
KTR 130		3	-0,08	2,85	
			-0,1		
KTR 150		3	-0,08	2,85	
			-0,1		





No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	NOTES, DIAGRAMS COMMENTS
2	Shield		Install shield into the Turbine Rotor					
3	Center Housing	Seal ring seat	Install center housing on the Turbine When installing the center housing, take care to prevent the seal ring from slipping to installation					
4	Journal Bearing	Journal bearing surface	Install journal bearing Apply sufficient amount of engine oil to the inside, outside and both sides of the journal Bearing prior to installation					

Measurement of Journal Bearing,Center Housing & Wheel Shaft

	Standard Size	Tolerance Shaft	Tolerance Hole	Repair limit Shaft	Repair limit Hole	Actual Shaft	Actual Hole
KTR 100							
Outside diameter of journal bearing	25	-0,04	0,02	24,91	25,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	17	-0,03	0,02	16,94	17,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET **TURBOCHARGER** ENGINE 6D170 SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

KTR 110

Outside diameter of journal bearing	25	-0,04	0,02	24,91	25,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	17	-0,03	0,02	16,94	17,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 130

Outside diameter of journal bearing	30	-0,04	0,02	29,91	30,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	20	-0,03	0,02	19,94	20,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

KTR 150

Outside diameter of journal bearing	30	-0,04	0,02	29,91	30,04		
Inner diameter of center housing		-0,06	0				
Inner diameter of Journal bearing	20	-0,03	0,02	19,94	20,05		
Outside diameter of wheel shaft		-0,05	0				
Bend of wheel shaft	Repair limit : 0.010 (Total indicated runout)						

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	SPV	
5	Thrust		• Install thrust collar • Make sure that the thrust collar is free from foreign material on both sides, apply engine oil (SAE #30) both sides of the collar, and install that collar with the match mark aligning with marking mark stamp on the turbine shaft. • Drive a dowel pin in the center housing.					
6	Thrust Bearing		• Install thrust bearing with the dowel pin fitted in position • Before installing the trust Bearing, apply engine oil (SAE #30) to both sides of the oil hole					

Thickness of thrust bearing

KTR 100	Standard size	Tolerance Width	Repair limit Width	Actual Width
	5	-0,08 -0,11	4,86	
KTR 110	Standard size	Tolerance Width	Repair limit Width	Actual Width
	5	-0,08 -0,11	4,86	
KTR 130	Standard size	Tolerance Width	Repair limit Width	Actual Width
	6	-0,08 -0,11	5,86	
KTR 150	Standard size	Tolerance Width	Repair limit Width	Actual Width
	6	-0,08 -0,11	5,86	

ASSEMBLY CHECK SHEET **TURBOCHARGER** ENGINE 6D170 SERIES						
	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN MEC	SIGN QA	SIGN SPV	NOTES, DIAGRAMS COMMENTS
7	Flinger		Fit seal ring into flinger					
			Apply engine oil to the sides of the seal ring					
			Install flinger into insert (back plate)					
		One seal ring	Let the gap of the seal ring on the flinger					
			Direct to the oil filler port on the center housing					
		Two seal ring	Let the gap of the seal ring in such away that the gap of each ring is positioned at 90° apart from from that of the Neighboring ring or at 45° apart from oil filler port on the center housing.					
8	Insert (Back Plate)		For KTR 100 , KTR 130 • Fit an O-ring and install insert					
			• Tighten the insert mounting bolts and then bend the lock washer Act. KTR 100 = kgm — KTR 100 Act. KTR 130 = kgm — KTR 130	1.25 ± 0.25 3.15 ± 0.35				
			For KTR 110,KTR 150 • Fit an O-ring on housing center					
			• Install back plate onto Turbine Rotor until contact the center housing and make sure that flinger with the match mark aligning with the marking mark stamp on the turbine shaft.					
			• Tighten the back plate mounting bolts and bend the lock washer. Act. KTR 110 = kgm — KTR 110 Act. KTR 150 = kgm — KTR 150	3.15 ± 0.35 6.75 ± 0.75				
9	Blower Impellers		Install blowers impellers onto the turbine rotor in the following manner immerse blower impeller in the oil (SAE #30) that is heated at 140° to 160° C for 5 to 20 minutes or immerse Blower					
			Impellers with hot air gun for 5-10 minutes.					
			Install the blower impellers into the turbine rotor, aligning the match mark with the match mark stamp on the turbine shaft.					 
			Apply engine oil to the inner face of the blower impeller and install the impeller without using excessive force					
			When tightening the nut, take care not allow any match marks to be out of alignment. Make sure that nut with the match mark aligning with the marking mark stamp on the turbine shaft and blower impeller.					 

PT SAPTAINDRA SEJATI
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ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE 6D170 SERIES						
	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED		SPEC.	SIGN MEC	SIGN QA	SIGN SPV	NOTES, DIAGRAMS COMMENTS
			Tightening torque :						
			Act. KTR 100 = kgm	KTR 100	4.25 ± 0.25				
			Act. KTR 110 = kgm	KTR 110	4.5 ± 0.25				
			Act. KTR 130 = kgm	KTR 130	8.5 ± 0.5				
			Act. KTR 150 = kgm	KTR 150	9.5 ± 0.5				
			Check End Play						
			Actual KTR 100 : = mm	KTR 100 Standard	0.06 to 0.11				
				Repair limit	0,18				
			Actual KTR 110 : = mm	KTR 110 Standard	0.08 to 0.13				
				Repair limit	0,18				
			Actual KTR 130 : = mm	KTR 130 Standard	0.08 to 0.13				
				Repair limit	0,18				
			Actual KTR 150 : = mm	KTR 150 Standard	0.08 to 0.13				
				Repair limit	0,18				
			Check Radial Play						
			Actual KTR 100 : = mm	KTR 100 Standard	0.20 to 0.44				
				Repair limit	0,6				
			Actual KTR 110 : = mm	KTR 110 Standard	0.25 to 0.44				
				Repair limit	0,6				
			Actual KTR 130 : = mm	KTR 130 Standard	0.25 to 0.43				
				Repair limit	0,6				
			Actual KTR 150 : = mm	KTR 150 Standard	0.25 to 0.43				
				Repair limit	0,6				
10	**Housing**	Turbine Housing	Take care not to damage the blower impeller and turbine impeller when installing the blower housing and turbine housing						 
			For KTR 110 , KTR 150 Install the Turbine housing into the center housing and the rotor assembly. Apply a thin coat of anti seiz compound to the flange section of turbine housing. Install lock plate, tighten mounting bolt. Apply a thin coat of anti seiz compound to the thread of bolt.						
			Tighten mounting bolt :						
			Act. KTR 110 = kgm	KTR 110	5.0 ± 0.5				
			Act. KTR 150 = kgm	KTR150	5.75 ± 0.25				
		Blower Impeller housing	Install blower housing, aligning it with the center housing and the rotor assembly. Apply a thin coat of Grease to the flange section.						
			For KTR 100 , KTR 130 Install the blower housing and turbine housing into the center housing and the rotor assembly.						
			Install V-Band clamp and tighten then torque of the V-Band locknut.						
			Act. KTR 100 = kgm	KTR 100	0.9 ± 0.1				
			Act. KTR 130 = kgm	KTR 130	0.9 ± 0.1				

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET **TURBOCHARGER** ENGINE 6D170 SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED		SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
						MEC	QA	SPV	
			Install V-Band clamp and tighten then torque of the V-Band locknut.						 
			Act. KTR 110 = kgm	KTR 110	0.9 ± 0.1				
			Act. KTR 150 = kgm	KTR150	0.9 ± 0.1				
			Measure the clearence in the radial direction in the following manner.						
			Depress the end of the rotor in the radial direction with a finger. Measure the clearence between the blower impeller and the housing by using filler gauge. Tollerance (Min.) :						
			Act. KTR 100 = mm	KTR 100	0,15				
			Act. KTR 110 = mm	KTR 110	0,2				
			Act. KTR 130 = mm	KTR 130	0,2				
			Act. KTR 150 = mm	KTR 150	0,2				

AFTER COMPLETING THE ASSEMBLY,ROTATE THE ROTOR ASSEMBLY WHILE DEPRESING IT LIGTHLY AND MAKE SURE THAT THE ROTOR CAN BE ROTATED SMOOTHLY WITHOUT BINDING.

NOTES :

ASSEMBLY BY

INSPECTION & APPROVAL BY :

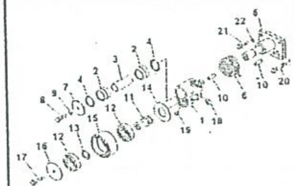
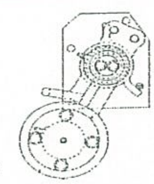
MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET TENSION PULLEY KOMATSU ENGINE 6D170-5			
JOB NUMBER		DOCUMENT NO	QR03-SA-009-15
ENGINE MODEL		PUBLISHED DATE	11-Jul-11
ENGINE S/N		REVISION NO	01
MACHINE MODEL		UNIT CODE	

NO	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
4	Assembly Pulley and Lever	- Install dowel pin (10) and spring (6) to shaft (5) - Install deowel pin (10) to bracket (1) - Install pulley and bracket assembly to shaft - Install cover lock (7) ,washer (9) and mounting bolt (8) then tighten tighten mounting bolts (17). Actual : kgm - Install cover (4), o-ring (9), snap ring (8), and plug (22)	6.7 ± 0.7 kgm				
5	Fitting grease and plug	- Install fitting grease (18) and relief valve (19), then tighten - Give all of surface pulley with grease to protect from rust					

⚠1 Change standard torque for pulley and lever assy by Mr.Hardi W / 11-07-2011 / Reman Standard

Notes :

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

RC/QA/EG/463	REV.01

PARTNER IN YOUR MINING NEEDS

PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

FUEL PUMP DRIVE ASSEMBLY

CHECK SHEET

KOMATSU ENGINE 6D170E-5

MACHINE MODEL

WO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
FUEL PUMP DRIVE

WO NUMBER		PART NAME		OCUMENT N	RC/QA/EG/463
ENGINE MODEL	6D170E-5	PART NUMBER		UBLISHED DAT	
ENGINE S/N		START DATE / TIME		REVISION NO	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	

ASSEMBLY BY	

NO	CHECK POINT	STANDAR	REPAIR LIMIT
1	Outside Diameter of shaft	**64.964~64.984**	–
	Inside Diameter of Bushing	65.066~64.996	–
	Clearance betwen OD shaft & ID bushing	0.012~0.102	–
2	Outside Diameter of Shaft	60.110~60.091	–
	Inside diameter of Flange	60.000~60.020	–
	Interference between OD shaft &ID Pulley	0.110~0.071	–

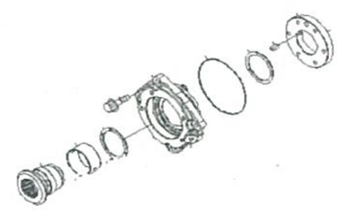
NO	CHECK POINT			ACTUAL			VISUAL CHECK (OK / NOT OK)
				MECHANIC	QA/GL	SPV	
1	OUTSIDE DIAMETER OF SHAFT	FRONT	X				
			Y				
		REAR	X				
			Y				
	INSIDE DIAMETER OF BUSHING	REAR	X				
			Y				
		FRONT	X				
			Y				
	CLEARANCE BETWEEN OUTSIDE DIAMETER SHAFT AND INSIDE DIAMETER BUSHING	REAR	MIN				
			MAX				
		FRONT	MIN				
			MAX				
2	OUTSIDE DIAMETER OF SHAFT	X1					
		Y1					
		X2					
		Y2					
	INSIDE DIAMETER OF FLANGE	X1					
		Y1					
		X2					
		Y2					
	CLEARANCE BETWEEN OUTSIDE DIAMETER SHAFT AND INSIDE DIAMETER PULLEY	1	MIN				
			MAX				
		2	MIN				
			MAX				

NO	PART	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN			VISUAL CHECK (OK or NOT OK)
					MEC	QA/GL	SPV	
1	ALL PART		CONFIRM THAT PART WICH HAVE ORDER DO NOT DAMAGE,SCRATCH AND COROSON					
2	HOUSING		CONFIRM THAT HOUSING FREE FROM DIRT,CRACK,COROSION AND DAMAGE CONFIRM THAT CONTACT BEARING FREE FROM CORROSION AND SRATCH CONFIRM THAT BOLT SEAT IS NOT DAMAGE SET SEAL SEAT AND O-RING POSITION					

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
FUEL PUMP DRIVE

WO NUMBER		PART NAME		DOCUMENT NO	RC/QA/EG/463
ENGINE MODEL	6D170E-5	PART NUMBER		PUBLISHED DATE	
ENGINE S/N		START DATE / TIME		REVISION NO	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	

NO	PART	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN			VISUAL CHECK (OK or NOT OK)
					MEC	QA/GL	SPV	
3	SHAFT		CONFIRM THAT SHAFT IS FREE FROM DIRT,CORROSON,CRACK,SCRATCH AND FITTING CONFIRM THAT CONTAC SEAL IS NOT SCARCH AND WORM					
	FLAGE		CONFIRM THAT FLANGE IS FREE FRM DIRT,CORROSION AND CRACK CONFIRM THAT FLANGE IS NOT FITTING AND DAMAGE CONFIRM THAT FLANGE IN CONTACT SHAFT IS NOT WEAR,SCRATCH AND FITTING					
5	ASSEMBLY BUSHING ON HOUSING		CONFIRM THAT BUSHING HOLE ITS OK INSTALL BUSHING IN HOUSING CONFIRM THAT BUSHING ISN'T DAMAGE					
6	ASSEMBLY SHAFT ON HOUSING		INSTALL BUSHING TO HOUSING INSTALL SHAFT IN HOUSINGAND LUBRICATED BY OIL INSTALL BEARING 2.PC IN SHAFT THAT WILL CONTACT WITH FLANGE INSTALL FLANGE IN SHAFT MEASURED ENDPLAY SUPPLYPUMP DRIVE GEAR endplay: mm	0.25 ~ 0.30 mm				

REVISION:

Change Standard Clearance Between OD Shaft and ID Bushing 13-07-2011/Drawing

Mechanic Assembly Comment

ASSEMBLY BY

INSPECTED & APROVED BY

adaro SERVICES

PT SAPTAINDRA SEJATI

PLANT REBUILD CENTER

FAN PULLEY

ASSEMBLY CHECK SHEET

ENGINE KOMATSU

6D170-5

MACHINE MODEL

WO NO:

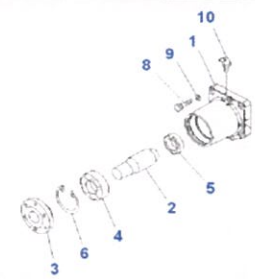
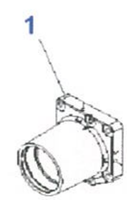
SUB ASSY SECTION

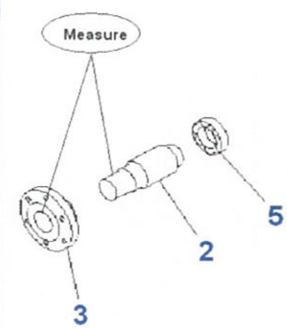
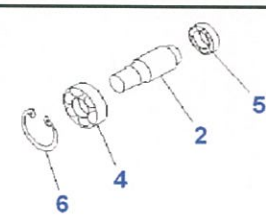
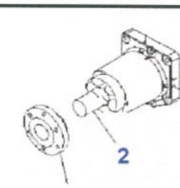
PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
FAN PULLEY DRIVE
KOMATSU ENGINE 6D170E-3

WO NUMBER		PART NAME		DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	30/09/2009
ENGINE S/N		START DATE / TIME		REVISION NO	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 2

ASSEMBLY BY		QUANTITY	

<table>
<tr><th rowspan="2">NO.</th><th rowspan="2">PARTS</th><th rowspan="2">CONDITIONS TO BE CHECKED</th><th rowspan="2">SPEC</th><th colspan="3">SIGN</th><th rowspan="2">REMARKS (SKETCH, PHOTOES, COMMENTS)</th></tr>
<tr><th>MEC</th><th>QA</th><th>SPV</th></tr>
<tr><td>1</td><td>All parts</td><td>- All parts must be free from dust, rush and dent
- All parts have been cleaning
- No crack, no scratch, no wear
- Always check the contact of the following parts before assembly.</td><td></td><td></td><td></td><td></td><td>
</td></tr>
<tr><td>2</td><td>Drive Housing</td><td>- Make sure drive housing is clean and free from damage.
- Inspect contact surface of drive housing
- No crack, no scratch, no wear</td><td></td><td></td><td></td><td></td><td>
</td></tr>
<tr><td>3</td><td>Drive Shaft Ass'y</td><td>- Check shaft from damage before assembly
- Measure out-side diameter of shaft (2) and inside-diameter of flange (3)

Actual Measure mm
<table>
<tr><th rowspan="2">ø</th><th colspan="2">DIMENSION</th></tr>
<tr><th>X-X</th><th>Y-Y</th></tr>
<tr><td>SHAFT</td><td></td><td></td></tr>
<tr><td>HOLE</td><td></td><td></td></tr>
<tr><td>INTERFERENCE</td><td></td><td></td></tr>
</table></td><td>STD :
UNIT = mm
50.046 ~ 50.030
50.000 ~ 50.030
0.014 ~ 0.046</td><td></td><td></td><td></td><td>
</td></tr>
<tr><td>4</td><td>Shaft and Bearing</td><td>- Coat the shaft contact bearing with grease
- Install bearing (5) to shaft (2)
- Install drive shaft ass'y to drive housing
- Install bearing (4) to shaft (2) and housing
*Using special tool to install bearing.
- Install snap ring (6) to lock bearing.</td><td>LG-7</td><td></td><td></td><td></td><td>
</td></tr>
<tr><td>5</td><td>Flange</td><td>- Check flange from damage before assembly
- Install flange (3) to shaft (2) and housing
- Coat contact surface with grease
*Using special tool.</td><td>LG-7</td><td></td><td></td><td></td><td>
</td></tr>
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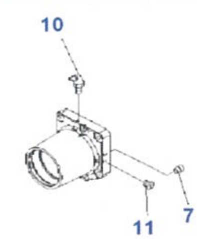
PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET
FAN PULLEY DRIVE
KOMATSU ENGINE 6D170E-3

WO NUMBER		PART NAME		DOCUMENT NO	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	30/09/2009
ENGINE S/N		START DATE / TIME		REVISION NO	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
				MEC	QA	SPV	
6	Plug	- Install plug (7), valve (11), then tighten - Instll plug (10), then tighten					
7	Fan drive ass'y	- Do greasing fan pulley pass to fitting grease - Coat fan drive contact surface with engine oil, then wrapping.					

Notes :

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

RC/QA/EG/050	REV. 01

PARTNER IN YOUR MINING NEEDS

PT.SAPTAINDRA SEJATI

CROSS HEAD SALVAGING PROCESS AND INSPECTION CHECK SHEET

KOMATSU ENGINE ALL SERIES

MACHINE MODEL

WO :

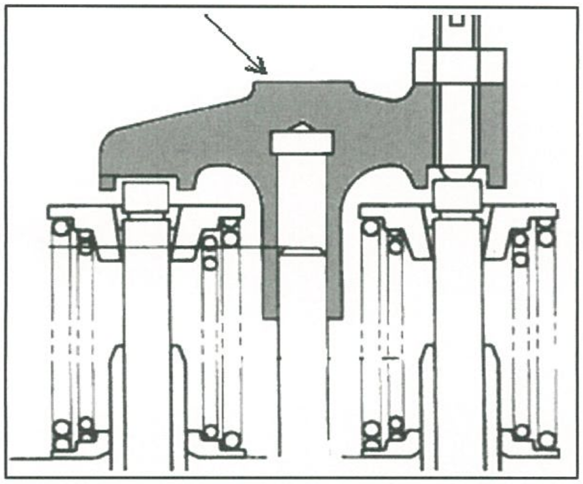
SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

SALVAGE CROSS HEAD PROCESS INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CROSS HEAD	DOCUMENT NO.	QR06-SAL-006
ENGINE MODEL		PART NUMBER		PUBL. DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

SKETCH DRAWING



MECHANIC NAME	

VISUAL CHECK	MECHANIC		QA	
	OK	NOT OK	OK	NOT OK
CRACK HAS BEEN CHECKED				
CHECK THREAD OF CROSSHEAD				
CHECK THREAD OF SCREW				
CHECK THREAD OF NUT				

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Polishing			
Measuring			
TOTAL			

Standard

Upper pad portion : Free from Wear

Physical Data

No.	Part Number	Part Name	Qty	New	Reuse
1		Cross head			
2		Screw			
3		Nut			

MECHANIC INSPECTION COMMENT :

INSPECTION BY

INSPECTION & APPROVED BY

MECHANIC

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION

QR08-SAL-015-03	REV. 04

PARTNER IN YOUR MINING NEEDS

PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ROCKER ARM SHAFT

MEASURING AND POLISHING CHECK SHEET

KOMATSU ENGINE 6D170E-5 SERIES

MACHINE MODEL

WO NO:

ENGINE SECTION

QR03-SA-008-03	REV.01

PARTNER IN YOUR MINING NEEDS

PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

ALTERNATOR DRIVE

ASSEMBLY CHECK SHEET

KOMATSU ENGINE 6D170E-5 SERIES

MACHINE MODEL

WO NO:

SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

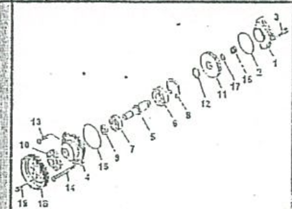
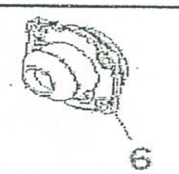
PARTNER IN YOUR MINING NEEDS

ASSEMBLY CHECK SHEET
ALTERNATOR DRIVE ENGINE 6D170-5

ISO

JOB NUMBER		PART NAME		DOCUMENT NO	QR03-SA-021-15
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	06-Dec-11
ENGINE S/N		START DATE / TIME		REVISION NO	01.
MACHINE MODEL		FINISH DATE / TIME		EKS UNIT CODE	

ASSEMBLY BY		QUANTITY	

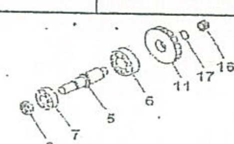
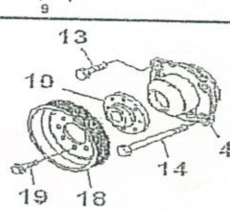
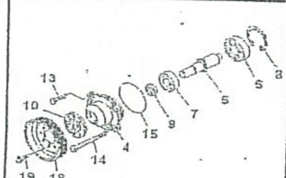
NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
1	All parts	- All parts must be free from dust, rush and dent - All parts have been cleaning - No crack, no scratch, no wear - Always check the contact of the following parts before assembly.					
2	Drive Housing	- Make sure drive housing is clean and free from damage. - Install bearing press fit to the drive housing *** Using special tool.**					
3	Drive Shaft Ass'y ⚠1	- Check shaft from damage before assembly - Measure out-side diameter of the shaft					

ACTUAL : Unit : mm

Ø	SHAFT CONTACT WITH GEAR	FLANGE	COUPLING GEAR
SHAFT			
HOLE			
Interference			

STANDARD : Unit : mm

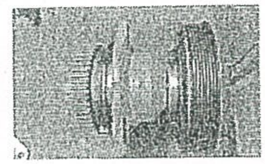
Ø	SHAFT CONTACT WITH GEAR	FLANGE	COUPLING GEAR
SHAFT	45.018 ~ 45.002	30.013 ~ 30.000	25.489 ~ 25.476
HOLE	44.941 ~ 44.966	29.938 ~ 29.959	25.400 ~ 25.421
Interference	0.077 ~ 0.036	0.062 ~ 0.028	0.089 ~ 0.055

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
3	Drive Shaft Ass'y	- Install bearing to the shaft - Install drive shaft ass'y in the housing					
4	Flange	- Install seal in the housing *** Using grease to install seal.** - Install press fit flange to the shaft *** Using special tool.**					
5	Assembly Shaft on Housing	- Install bearing to shaft - Install bearing 1 pcs in shaft which will contact with housing. - Install Shaft in housing and lubricated by oil. - Install bearing 1 pcs in shaft that will contact with gear. - Install snap ring in shaft.					

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
ALTERNATOR DRIVE ENGINE 6D170-5

JOB NUMBER		PART NAME		DOCUMENT NO	QR03-SA-021-15
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	06-Dec-11
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		EKS UNIT CODE	

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN MEC	SIGN QA	SIGN SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
6	Fuel Pump drive ass'y	- Coat fuel pump drive contact surface with engine oil, then wrapping.					

⚠1 Change format checking Shaft contact with Gear and Gear Coupling / 06-12-2011 / Reman Standard

Notes :

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)

QR03-SA-008-03	REV.01

PARTNER IN YOUR MINING NEEDS

PT.SAPTAINDRA SEJATI

PLANT REBUILD CENTER

FAN PULLEY DRIVE ASSEMBLY

CHECK SHEET

KOMATSU ENGINE SAA6D170E-5

MACHINE MODEL

WO NO:

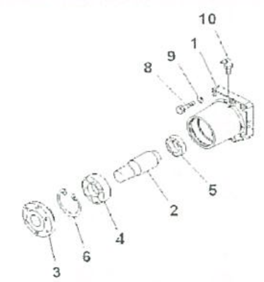
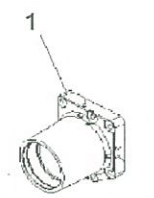
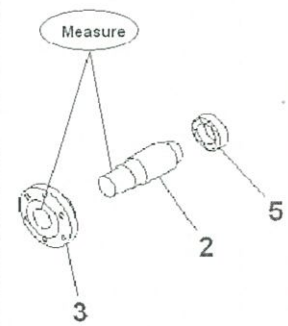
SUB ASSY SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
FAN PULLEY DRIVE
KOMATSU ENGINE 6D170E-5

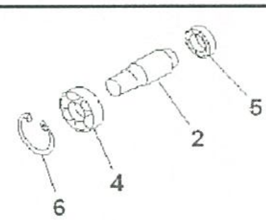
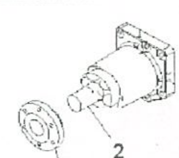
WO NUMBER		PART NAME		Eks UNIT CODE	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	30/09/2009
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO	1 of 2

ASSEMBLY BY		QUANTITY	

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
				MEC	QA	SPV	
1	All parts	- All parts must be free from dust, rush and dent - All parts have been cleaning - No crack, no scratch, no wear - Always check the contact of the following parts before assembly.					
2	Drive Housing	- Make sure drive housing is clean and free from damage. - Inspect contact surface of drive housing - No crack, no scratch, no wear					
3	Drive Shaft Ass'y	- Check shaft from damage before assembly - Measure out-side diameter of shaft (2) and inside-diameter of flange (3)	STD : UNIT = mm 50.046 ~ 50.030 50.000 ~ 50.030 0.014 ~ 0.046				

Actual Measure mm

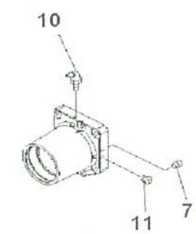
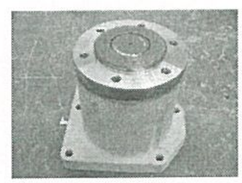
Ø	DIMENSION	
	X-X	Y-Y
SHAFT		
HOLE		
INTERFE RENCE		

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	MEC	QA	SPV	REMARKS (SKETCH, PHOTOES, COMMENTS)
4	Shaft and Bearing	- Coat the shaft contact bearing with grease - Install bearing (5) to shaft (2) - Install drive shaft ass'y to drive housing - Install bearing (4) to shaft (2) and housing ***Using special tool to install bearing.** - Install snap ring (6) to lock bearing.	LG-7				
5	Flange	- Check flange from damage before assembly - Install flange (3) to shaft (2) and housing - Coat contact surface with grease ***Using special tool.**	LG-7				

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET
FAN PULLEY DRIVE
KOMATSU ENGINE 6D170E-5

WO NUMBER		PART NAME		Eks UNIT CODE	
ENGINE MODEL		PART NUMBER		PUBLISHED DATE	30/09/2009
ENGINE S/N		START DATE / TIME		REVISION NO	01
MACHINE MODEL		FINISH DATE / TIME		DOCUMENT NO	2 of 2

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
				MEC	QA	SPV	
6	Plug	- Install plug (7), valve (11), then tighten - Instll plug (10), then tighten					
7	Fan drive ass'y	- Do greasing fan pulley pass to fitting grease - Coat fan drive contact surface with engine oil, then wrapping.					

Notes :

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)